12+ page PDF handbook

PRECISION POLLINATION HANDBOOK

BeeYield's operating model for crop deployment, bloom timing, colony grading, and frames per acre.

Who this is for
Growers, pollination coordinators, field supervisors, and professional beekeepers working with managed crop pollination.

Guide summary
This handbook explains precision pollination as BeeYield practises it: matching crop demand to colony strength, bloom timing, field placement, and live verification. It also documents the frames-per-acre model used in the BeeYield dashboard so growers understand what the metric means and how deployment decisions are made.

BeeYield brand standard: every page of this PDF carries BeeYield branding, page numbering, and a repeatable working format so the guide can be used in the yard, classroom, or packing room.

How to use this guide

Read the main sections in order, then use the worksheets and tables as operational tools. This document is written for field use, so the recommended action is always more important than memorising definitions.

12 frames	**24 FPA**	**48 FPA**	**3**
Grade A colony	Mango target	Onion target	Core field factors

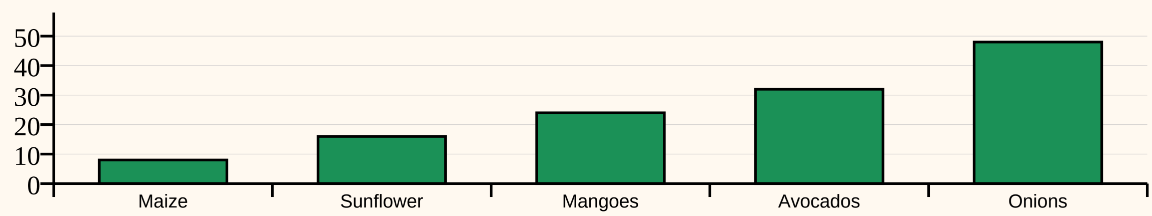
Learning outcomes

- Understand the difference between paying for hive boxes and paying for pollination force.
- Calculate frames per acre and required hive numbers from colony strength and acreage.
- Use crop-specific BeeYield target ranges to brief teams before deployment.
- Interpret field signals such as bloom intensity, weather risk, and monitored hive performance.

BeeYield Crop Profiles in the Current FPA Model

Maize	4	8	12
Sunflower	12	16	24
Mangoes	16	24	32
Avocados	20	32	40
Beans	8	12	18
Oranges	8	20	28
Tomatoes	8	12	18
Onions	32	48	60
Sisal	2	4	8

Recommended FPA by crop profile



Full-width implementation infographic

Use this sequence page as a one-glance map of the guide. It is designed for quick team briefings and field refreshers before work starts.

PRECISION POLLINATION HANDBOOK workflow map

A visual sequence of how the guide should be applied in the field or business workflow.

Step 1
Precision pollination is o

Step 2
Frames per acre is the sim

Step 3
Crop profiles set the dema

Step 4
Colony grading decides whe

Precision pollination is
Measure force, not just box count.

Frames per acre is the s
FPA = total frames of bees / total acr

Crop profiles set the de
Use crop profiles as planning defaults

12 frames
Grade A colony

24 FPA
Mango target

48 FPA
Onion target

3
Core field factors

1. Precision pollination is operational discipline, not just a slogan

Traditional pollination agreements often pay for hive counts without proving whether those hives delivered enough foraging force when bloom was actually open. Precision pollination starts by asking a harder question: how much pollination capacity reached the crop, at the right time, under the field conditions that existed that week?

BeeYield treats pollination as a measurable service. Colony strength, acreage, crop profile, bloom timing, placement geometry, weather, and in-hive monitoring all shape the final result. This makes the discussion more demanding, but it also makes the work more honest. The grower can see why one field needs reinforcement while another is already well covered.

In practice, precision pollination means fewer assumptions and faster correction. Instead of discovering weak coverage at fruit set, the team reads the indicators during bloom and adjusts deployment before the loss is locked in.

Field actions

- Measure force, not just box count.
- Match deployment to the crop's real bloom window.
- Verify performance while correction is still possible.
- Keep grower communication tied to evidence from the field.

2. Frames per acre is the simplest shared language for strength

BeeYield's current model uses frames per acre, or FPA, as the core density metric. At its simplest, FPA equals the total frames of bees deployed divided by the total acreage covered. A field receiving 480 frames across 20 acres is running at 24 FPA. This is easier to compare than raw hive count because two apiaries can deliver very different force with the same number of boxes.

The practical planning formula is just as direct: required hives = acres multiplied by target FPA, divided by the average frames per hive. If a 20 acre mango block is targeted at 24 FPA and the team is sending Grade A colonies averaging 12 frames, the starting requirement is 40 hives. If the same job is staffed with weaker colonies, the hive count must rise or the grower is under-covered from the start.

BeeYield also tracks effective FPA. That is the planning density after bloom intensity, forage competition, and weather risk trim the real working force. The concept matters because identical box counts can perform very differently in a clean peak bloom versus a windy week with scattered flowering.

Field actions

- FPA = total frames of bees / total acres.
- Required hives = (acres x target FPA) / average frames per hive.
- Effective FPA is lower when weather, weak bloom, or competition reduce field performance.
- Use FPA to brief both growers and bee teams in the same units.

3. Crop profiles set the demand side of the model

Not every crop asks the same thing from bees. Some blocks need a short burst of high pressure during a concentrated bloom, while others need a more moderate but consistent presence. The BeeYield crop profiles in the live calculator reflect that difference by setting minimum, recommended, and upper operating ranges.

For example, the current model treats onions as a high-density pollination job because seed production is unforgiving when bee traffic is light. Sisal sits at the other end of the range because it is managed more as part of ecological support and forage planning than a high-demand fruit set contract. Mangoes, avocados, sunflower, beans, oranges, and tomatoes sit between those extremes with different target bands.

These values should be presented as BeeYield operational defaults, not universal law. A crop profile is a starting point that still has to be read against bloom quality, competing forage, weather, terrain, and the quality of the colonies on site.

Field actions

- Use crop profiles as planning defaults, then adjust for field reality.
- High-value seed or fruit contracts justify tighter verification and stronger colonies.
- A field with uneven bloom may need better placement before it needs more hives.
- Document every deviation from the default so future pricing improves.

4. Colony grading decides whether the maths means anything

Frames per acre only works if the frame count reflects live colony strength. BeeYield's calculator uses colony grade to estimate the average frames per hive. In the current model, Grade A colonies are planned at 12 frames, Grade B at 8, and Grade C at 6. The quality of the deployment therefore depends on honest grading before trucks leave the yard.

A disciplined grading process checks adult bee coverage, brood pattern, queen-right status, food margin, and temperament. Weak hives hidden inside a strong pallet distort the contract and create avoidable conflict in the field. The cost of replacing non-performing colonies during bloom is far higher than the cost of grading properly before dispatch.

Where sensors are installed, monitored temperature stability, weight movement, and acoustic confidence can strengthen the grading decision. The point is not to replace beekeeper judgement, but to give it a second signal that is harder to argue with when the grower asks for proof.

Field actions

- Grade colonies before loading, not after a complaint.
- Keep grading criteria written and consistent across crews.
- Use stronger colonies on premium contracts and shorter bloom windows.
- Replace non-performing hives fast enough to preserve the bloom window.

5. Placement and verification keep density from becoming theory

A perfect FPA value can still fail if colonies are placed where bees waste flight time crossing barriers, competing with richer forage outside the block, or clustering too far from the main bloom. Precision pollination therefore treats mapping and spacing as part of the product. Hives should be placed so bees enter the crop quickly and coverage is distributed across the field, not concentrated at convenient roadside points.

Verification is the second half of the job. Delivery time, GPS location, colony count, crop stage, bloom notes, and sensor IDs should be logged when hives are placed. During bloom, the team should compare expected flight conditions with what monitored colonies and field observations are showing. A gap discovered on day three is fixable. A gap discovered after poor fruit set is a postmortem.

Growers gain confidence when the reporting language is plain: target, actual, risk, and action. Long reports do not replace clear evidence.

Field actions

- Map drop points before trucks move.
- Use bloom observations to confirm that bees are working the intended crop.
- Log delivery, replacements, and removals the same day.
- Report in decision terms: what changed, why, and what will be done next.

6. The commercial value is accountability

Growers do not buy pollination mathematics for its own sake. They buy reduced uncertainty around yield, fruit set, and contract performance. The business value of precision pollination is that it makes beekeeper and grower responsibilities visible. If bloom was weak, weather was hostile, or colonies were below grade, the discussion moves from accusation to evidence.

For BeeYield, the model also improves pricing. Stronger colonies, tighter verification, and premium reporting can be priced as a better service because the service is defined clearly. Teams can compare jobs after the season and see where margins were lost through under-grading, poor logistics, or unplanned replacements.

Precision pollination is therefore not just a technical layer over beekeeping. It is a management system that aligns biological performance, field execution, and commercial trust.

Field actions

- Translate technical metrics into grower confidence.
- Use post-season reviews to improve the next contract.
- Price premium service against evidence and responsiveness, not marketing claims alone.
- Treat accountability as a core deliverable.

Glossary

FPA	Frames per acre, the number of active frames of bees allocated across one acre.
Target FPA	The BeeYield operating target used to size deployments for a crop and bloom stage.
Effective FPA	Frames per acre after adjusting for bloom, forage competition, and weather risk.
Colony grade	A practical class for hive strength used when planning or pricing pollination work.
Bloom intensity	How strong and uniform the current flowering window is across the field.
Coverage gap	The shortfall between effective pollination force and the target requirement.

Quick reference	Use it when
Precision pollination is operational discipline, not just a slogan	Measure force, not just box count.
Frames per acre is the simplest shared language for strength	FPA = total frames of bees / total acres.
Crop profiles set the demand side of the model	Use crop profiles as planning defaults, then adjust for field reality.
Colony grading decides whether the maths means anything	Grade colonies before loading, not after a complaint.

Pollination Deployment Planner

Worksheet purpose

List crop, acreage, target FPA, average frames per hive, bloom intensity, forage competition, and the final hive count committed.

Date / block __________________	**Operator** __________________	**Priority** Low / Med / High
Main observation ______________________________ ____	**Decision taken** ______________________________ ____	**Follow-up due** ______________________________ ____

Working notes

Grower Verification Sheet

Worksheet purpose

Use this sheet to log delivery time, mapped drop points, colony grade, sensor IDs, weather, and corrective actions during bloom.

Date / block ________________	**Operator** ________________	**Priority** Low / Med / High
Main observation ____________________________	**Decision taken** ____________________________	**Follow-up due** ____________________________

Working notes

References

This guide was written from BeeYield operating practice and the following external references. Review them when adapting the guide to a new crop, region, or compliance requirement.

Pollinator Partnership - Pollination and Production
https://pollinator.org/pollinator.org/assets/generalFiles/P2-Ag_PSC_02-28-2025.pdf

USDA ARS - Monitoring colony phenology using within-day variability in continuous weight and temperature of honey bee hives
https://www.ars.usda.gov/research/publications/publication/?seqNo115=313374

FAO / Bees for Development - Beekeeping and sustainability
https://www.fao.org/family-farming/detail/en/c/1755592/

BeeYield internal crop profiles and pollination calculator logic
Internal BeeYield model documented in src/lib/apicultureModels.ts and src/lib/pollinationCalculations.ts